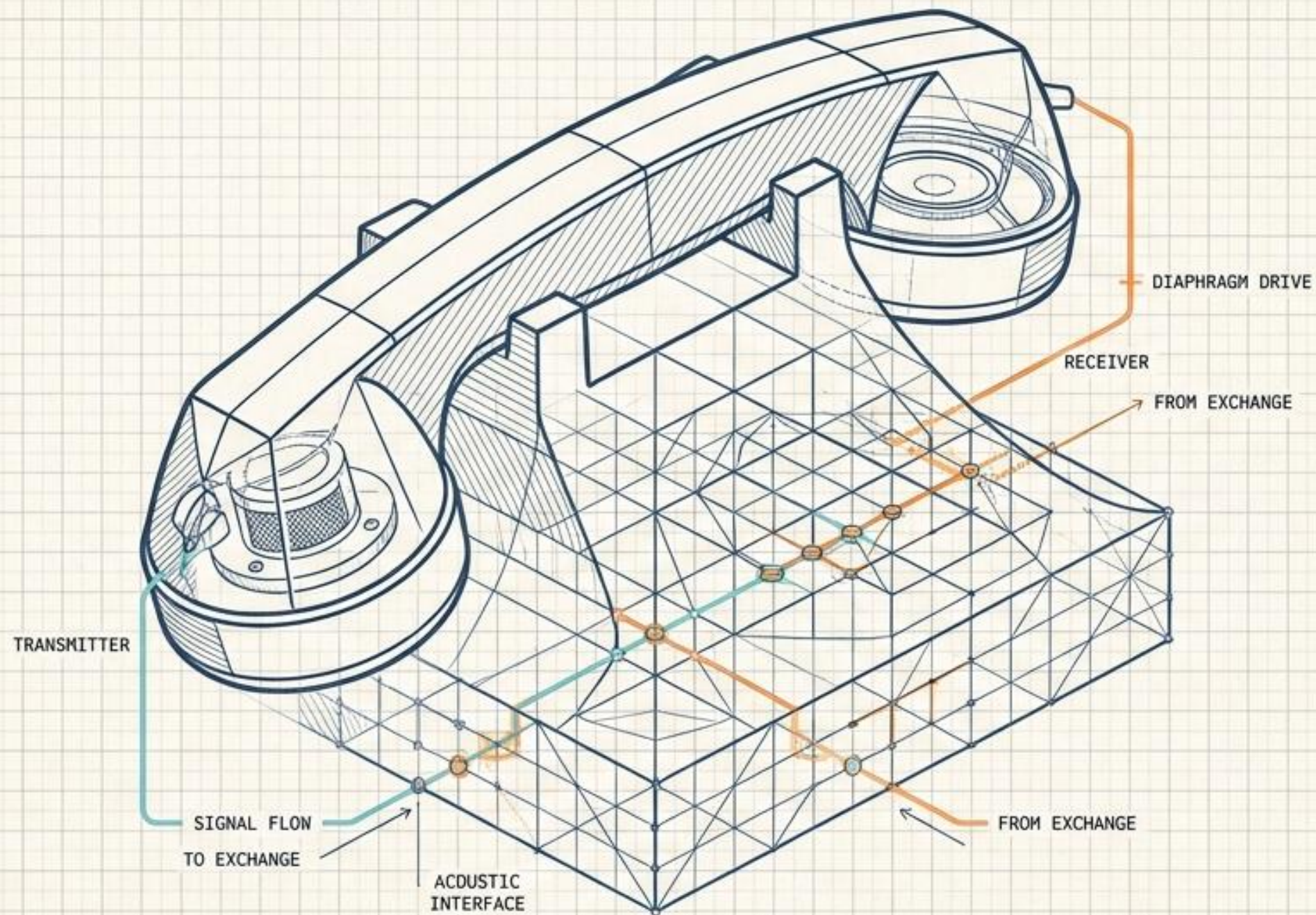
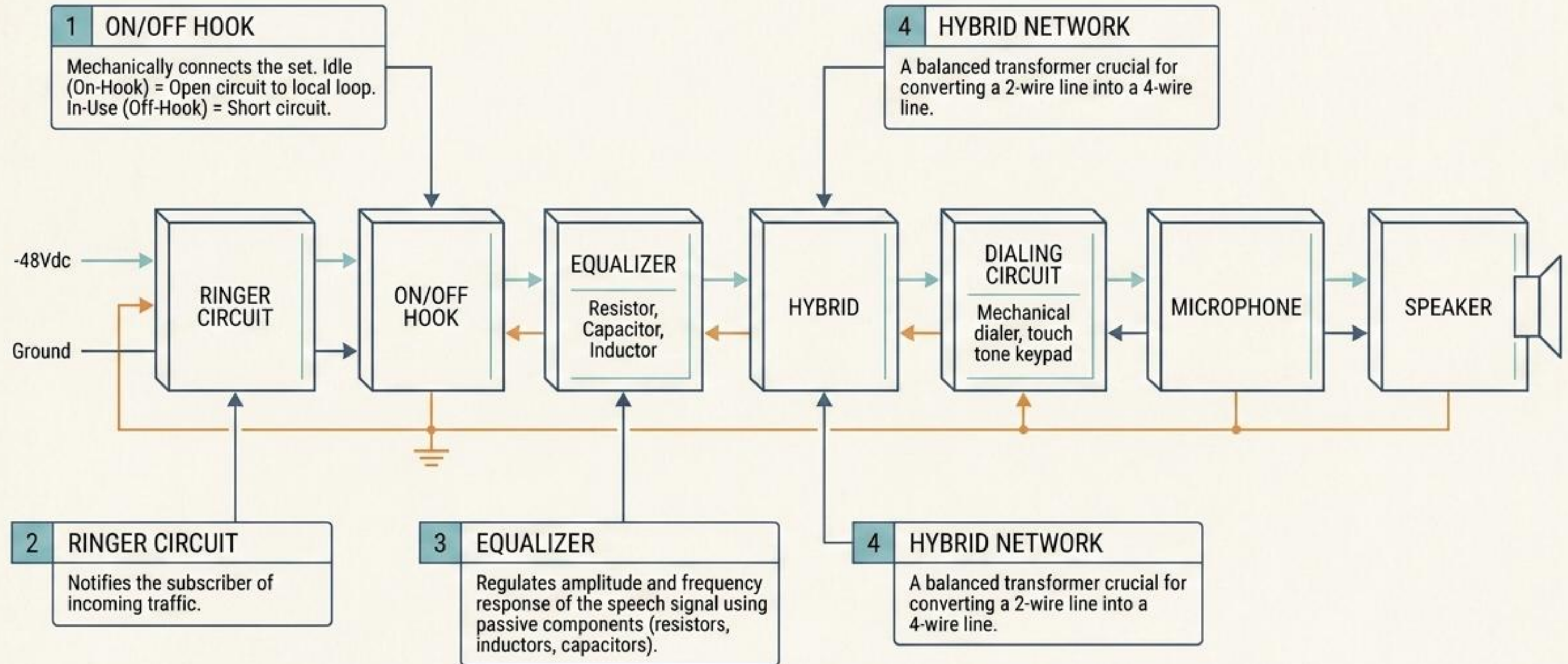


The Anatomy of a Telephone System

From Acoustic Transducers to Network Architecture

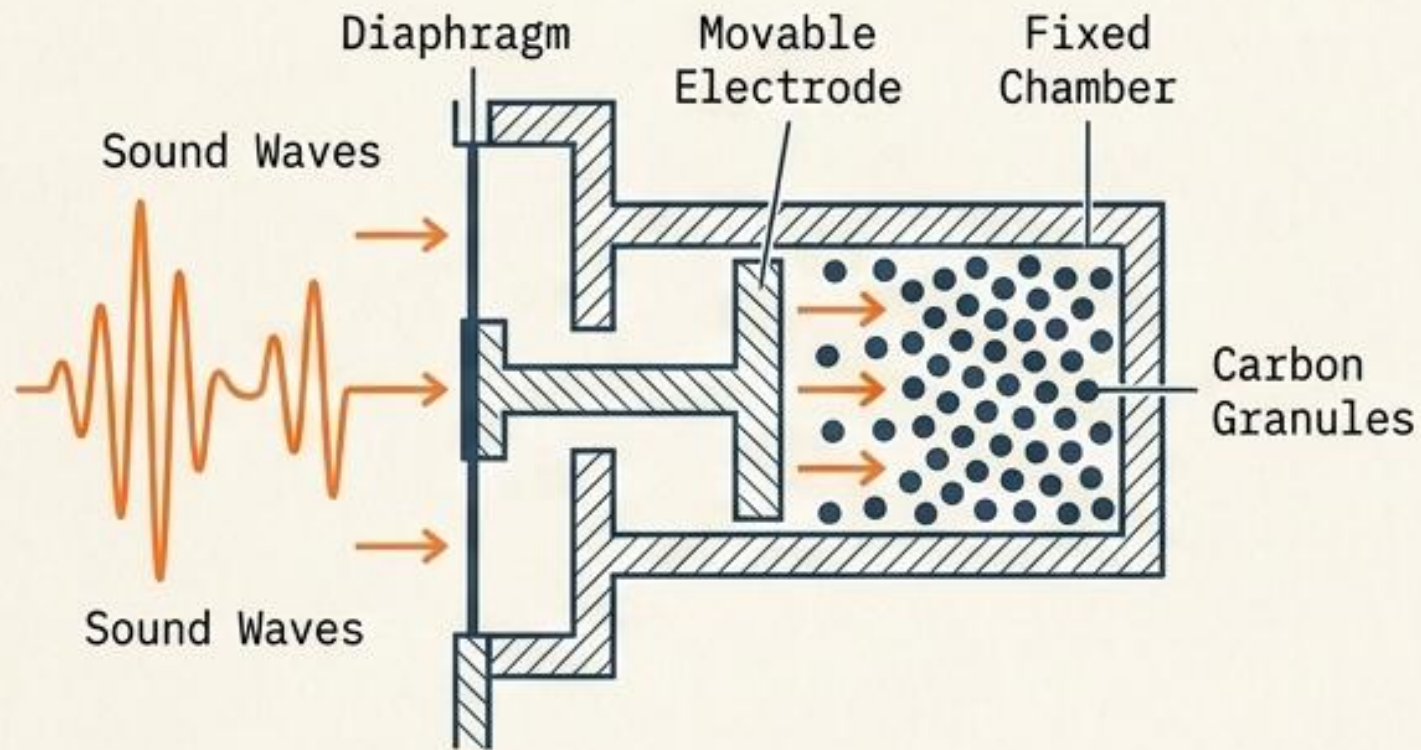


Inside the Telephone Set

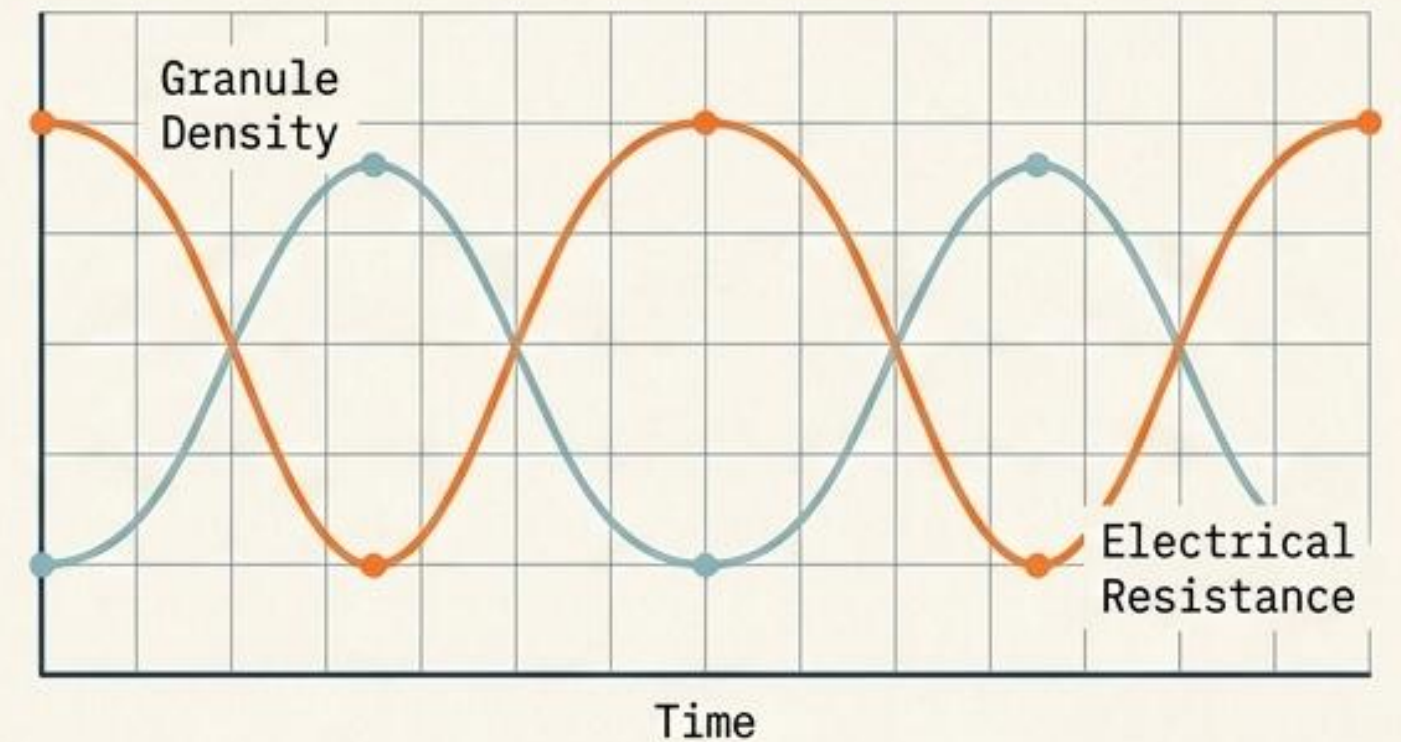


The Transducer: Carbon Microphone Mechanics

Frame A: Mechanical



Frame B: Electrical



The Transducer Principle

Carbon granules conduct electricity. Their electrical resistance is directly proportional to how densely they are packed together.

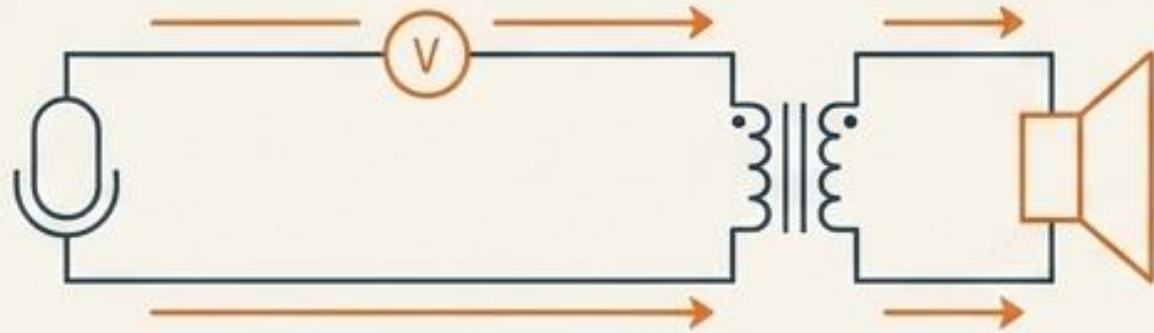
The Chain Reaction

Sound waves impinge on the diaphragm -> Diaphragm vibrates -> Granules compress and expand -> Electrical resistance changes synchronously with the acoustic sound.

Circuit Evolution: Simplex vs. Half-Duplex



One-Way Traffic

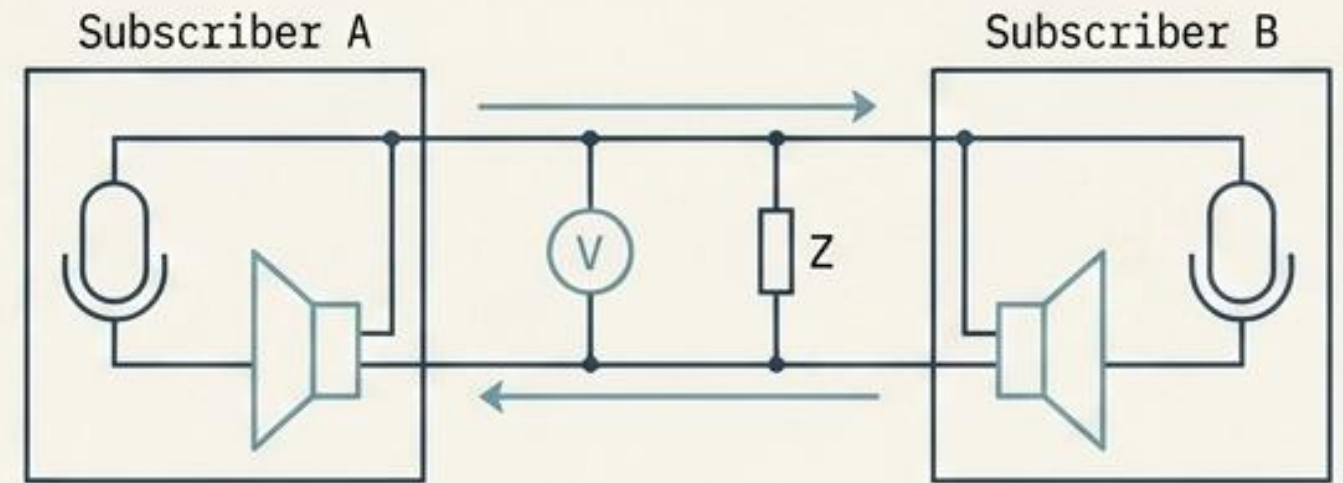


Simplex Limitations

Basic transmission requires an isolated transmitter and receiver loop, allowing communication in only one direction.



Shared Two-Way Traffic



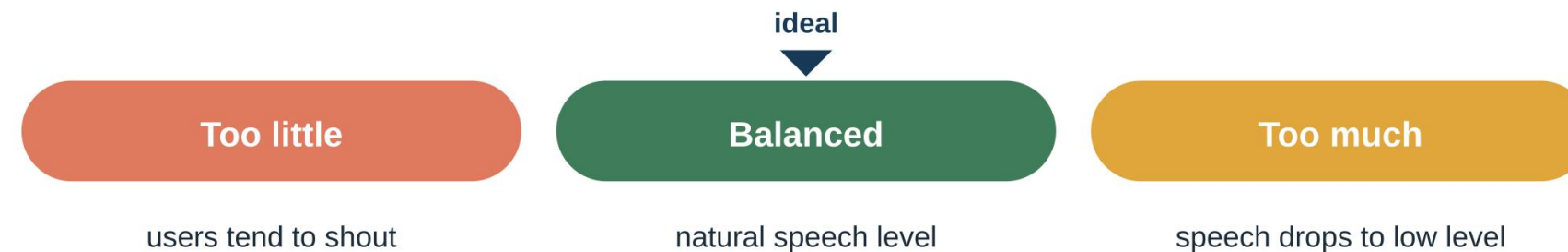
The Half-Duplex Solution

To achieve two-way (but not simultaneous) communication, both terminals must house a transmitter and receiver on a shared circuit.

The Unintended Consequence

Because the circuit is shared, the speech of Subscriber A is heard not only by Subscriber B, but also by Subscriber A's own earphone. This is known as Sidetone.

A Small Amount Is Necessary



Sidetone is the speaker's own voice fed back into their earphone — and the level has to be just right.



Too little → people shout.

With no feedback the line feels dead, so callers raise their voices.

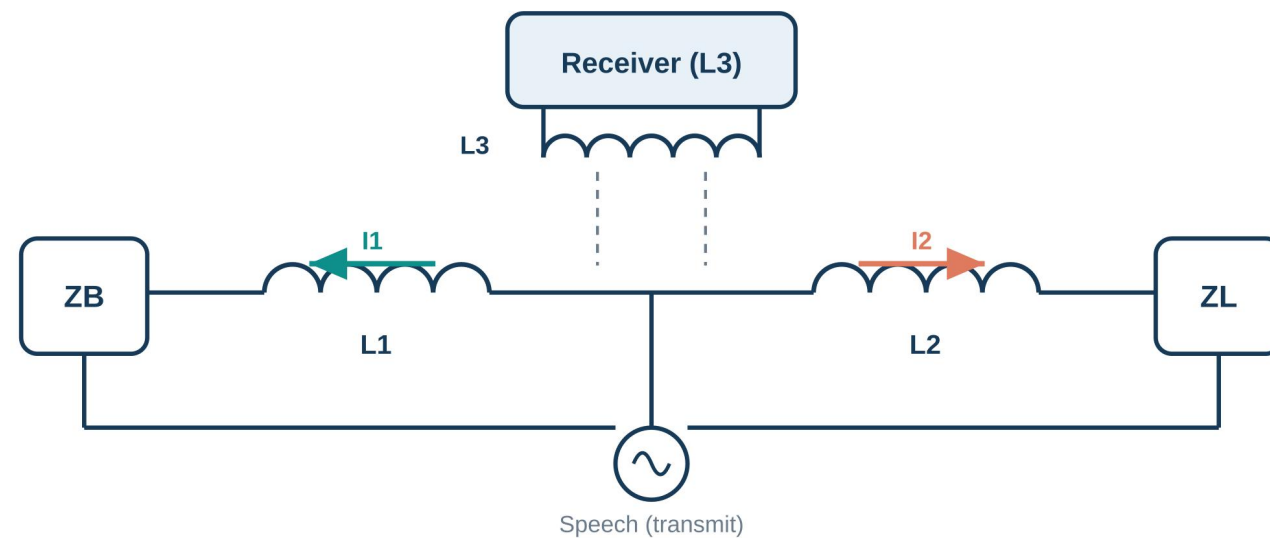


Too much → speech drops.

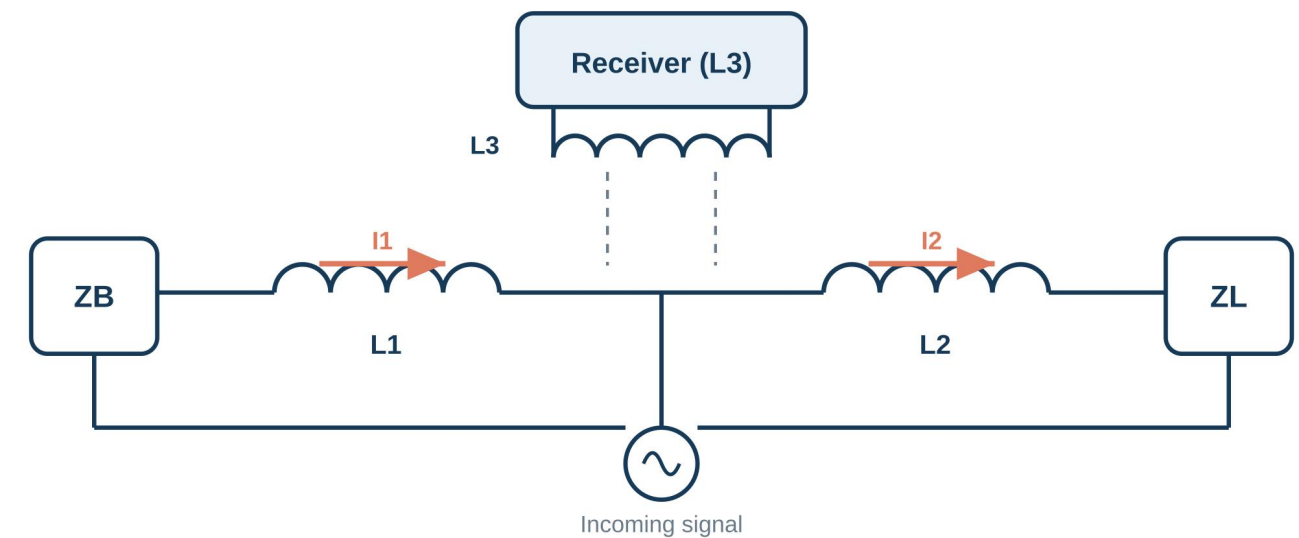
Hearing too much of themselves, callers lower their voices to a low level.

Anti-Sidetone Circuit — Transmit vs. Receive

Transmitter — currents oppose → low sidetone



Receiver — current adds → high sidetone



Transmitting — low sidetone

Current through L1 and L2 flows in **opposite directions**, so only a small resultant current is induced in L3. The speaker hears little of their own voice.

Receiving — high sidetone

Current through the coils flows in the **same direction**; the resultant current adds and is induced in L3 — so the incoming caller is heard clearly.

Sidetone vs. Echo



Sidetone	Echo
The speaker's own voice fed back to their own earphone.	Audio reflected back to the transmitter due to impedance mismatch.
A small amount is necessary.	Always unnecessary and annoying.
Anti-sidetone circuits reduce undesired sidetone.	Echo suppressors / cancellers are used to overcome it.
Too much makes users lower their volume.	Too much echo is highly distracting.

Switching System Parameters



Inlets / Outlets

Number of input and output terminals of the switch.



Switching Elements (S)

Number of crosspoints in the switching matrix.



Cost (C)

Cs = per element · Ce = common control · Coh = hardware

$$C = S \cdot C_s + C_e + C_{oh}$$



Switching Capacity (SC)

Maximum number of simultaneous connections.



Traffic-Handling Capacity

Ratio of switching capacity to maximum theoretical load.

$$TC = SC \div (N/2)$$



Equipment Utilization (EUF)

Elements in use vs. total — higher is better.

$$EUF = SE \text{ in use} \div \text{total SE}$$

Cost-Capacity Index & Folded Network



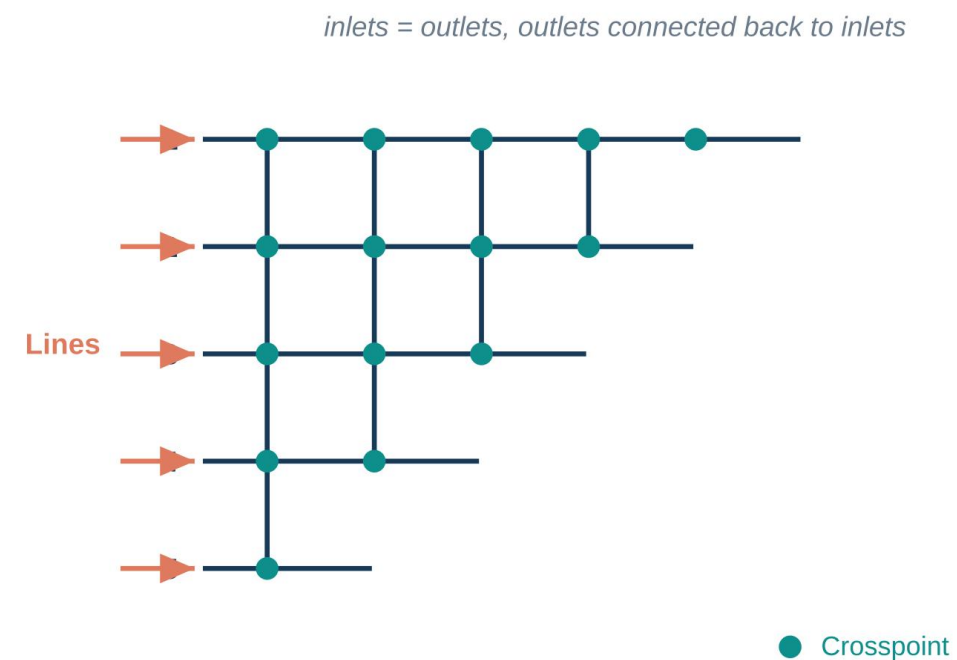
Cost-Capacity Index (CCI)

$$\text{CCI} = \text{Switching capacity} \div \text{Cost per line}$$

Higher CCI $\Rightarrow$ better design.

Folded Network

- A symmetric network in which the number of inlets equals the number of outlets.
- The outlets are connected back to the inlets, giving a triangular crosspoint array.



Switching System Parameters & Metrics

Traffic Handling Capacity (TC)

Ratio of switching capacity to maximum theoretical load. For N subscribers, max connections is N/2.

$$TC = \frac{2 \times SC \text{ }{N \text{ $$

Equipment Utilization Factor (EUF)

Measure of active hardware efficiency. Higher EUF indicates proper utilization of equipment.

$$\frac{\text{Numerator: } \text{ \text{ Switch Elements in Operation}}{\text{Denominator: Total Available Switch Elements}}$$

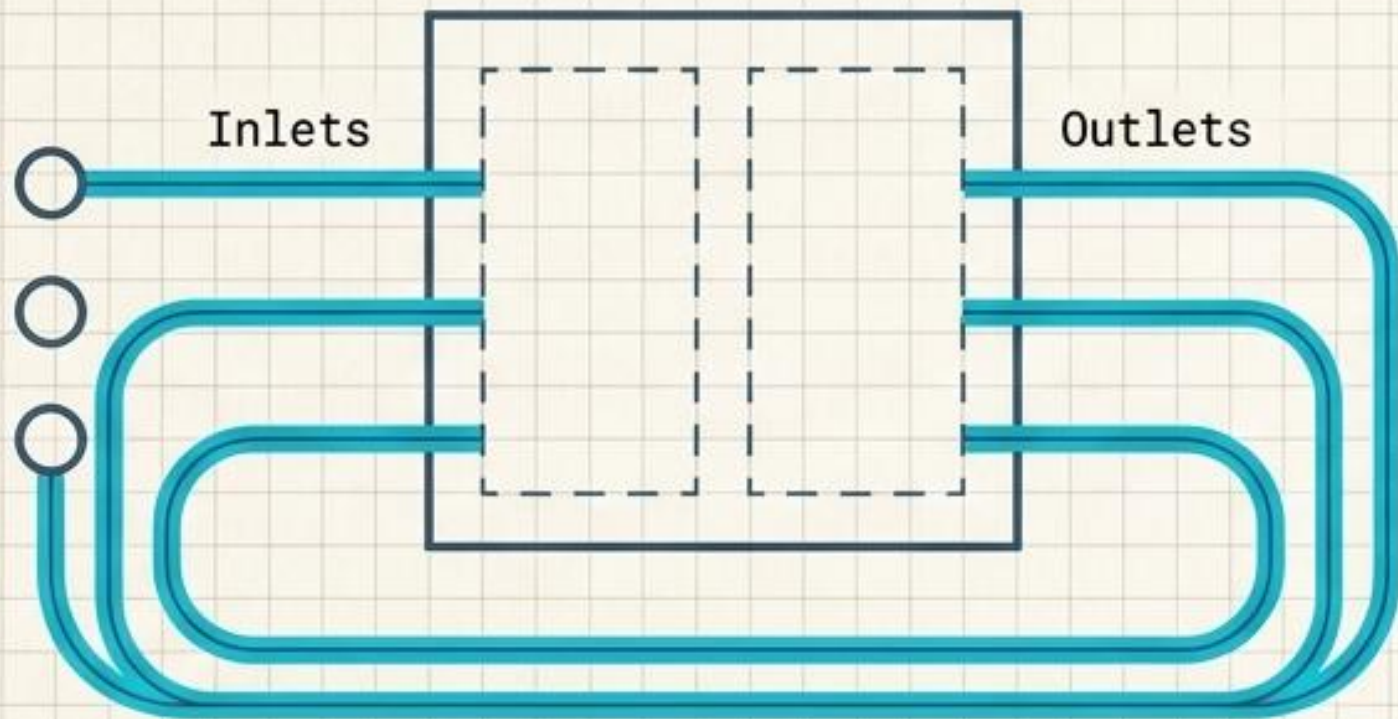
Cost Capacity Index (CCI)

Ratio of switching capacity to the cost per subscriber line. A higher value indicates a better engineering design.

$$CCI = \frac{N \text{  \times SC \text{ }{C \text{ $$

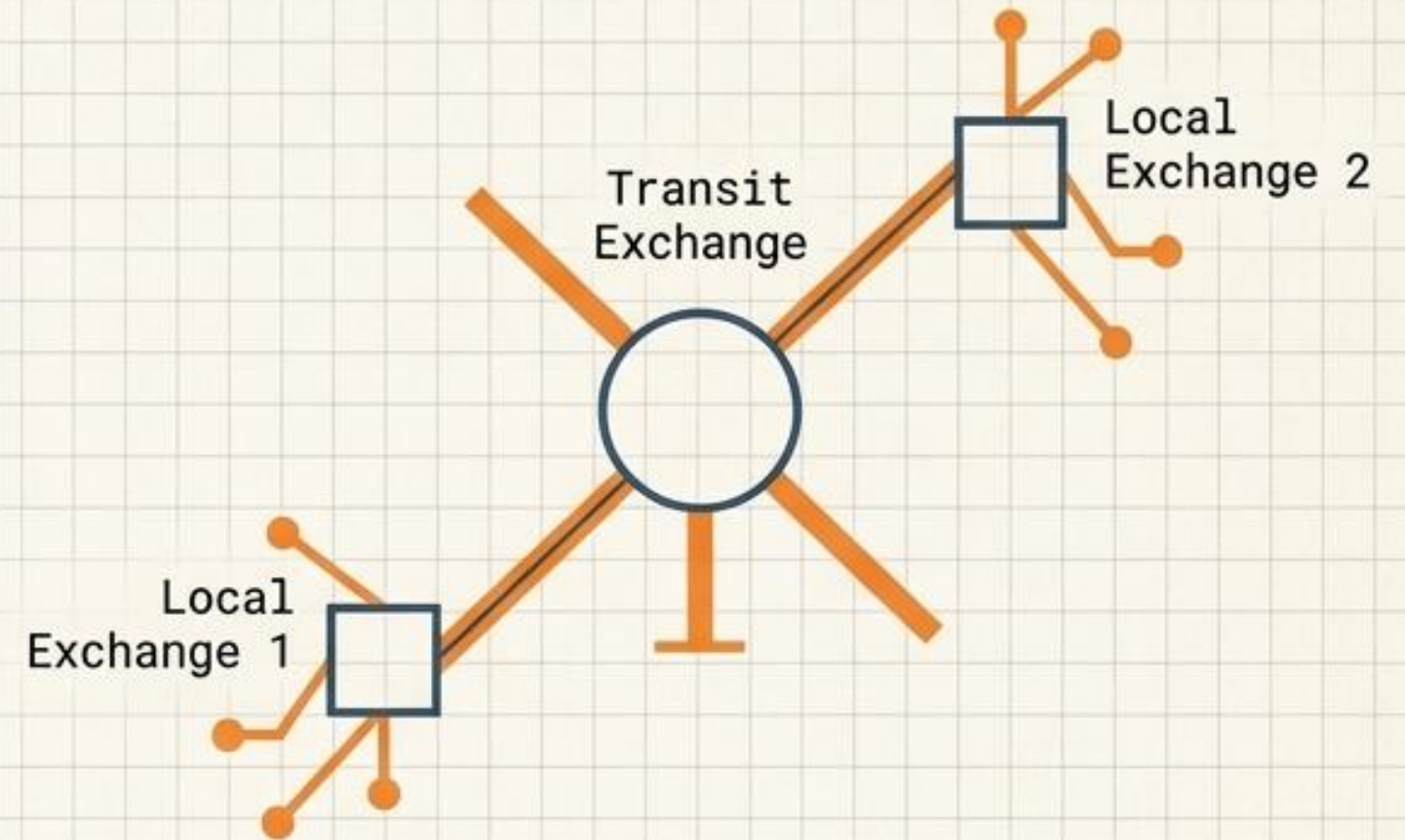
Total Cost (C) = (Switch Elements x Cost per Element)
+ Common Control Cost + Hardware Cost

Network Topologies



The Folded Network

A symmetric local network where the number of inlets perfectly equals the number of outlets, and all traffic remains internal to the closed loop.

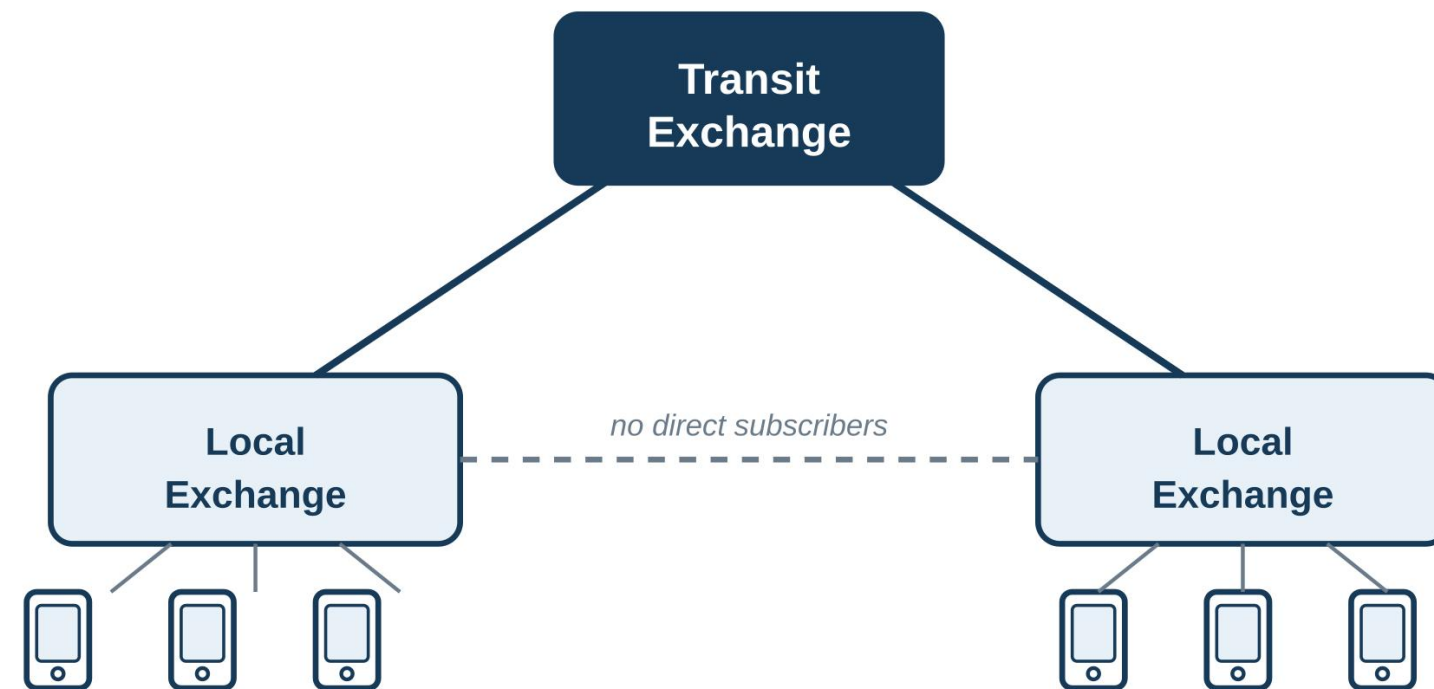


The Transit Exchange

A specialized central hub acting as a bridge exclusively between two local exchanges. It does not serve subscribers directly; it is engineered for heavy traffic routing in urban areas.

Transit Exchange & Intelligent Network

Transit Exchange



A switch that connects circuits between other exchanges but does not serve subscribers directly — used to handle traffic, e.g. in urban areas.



Intelligent Network

Separates service logic from the switching equipment, so new services can be added without redesigning switches.

Services it enables



Prepaid system



Calling card

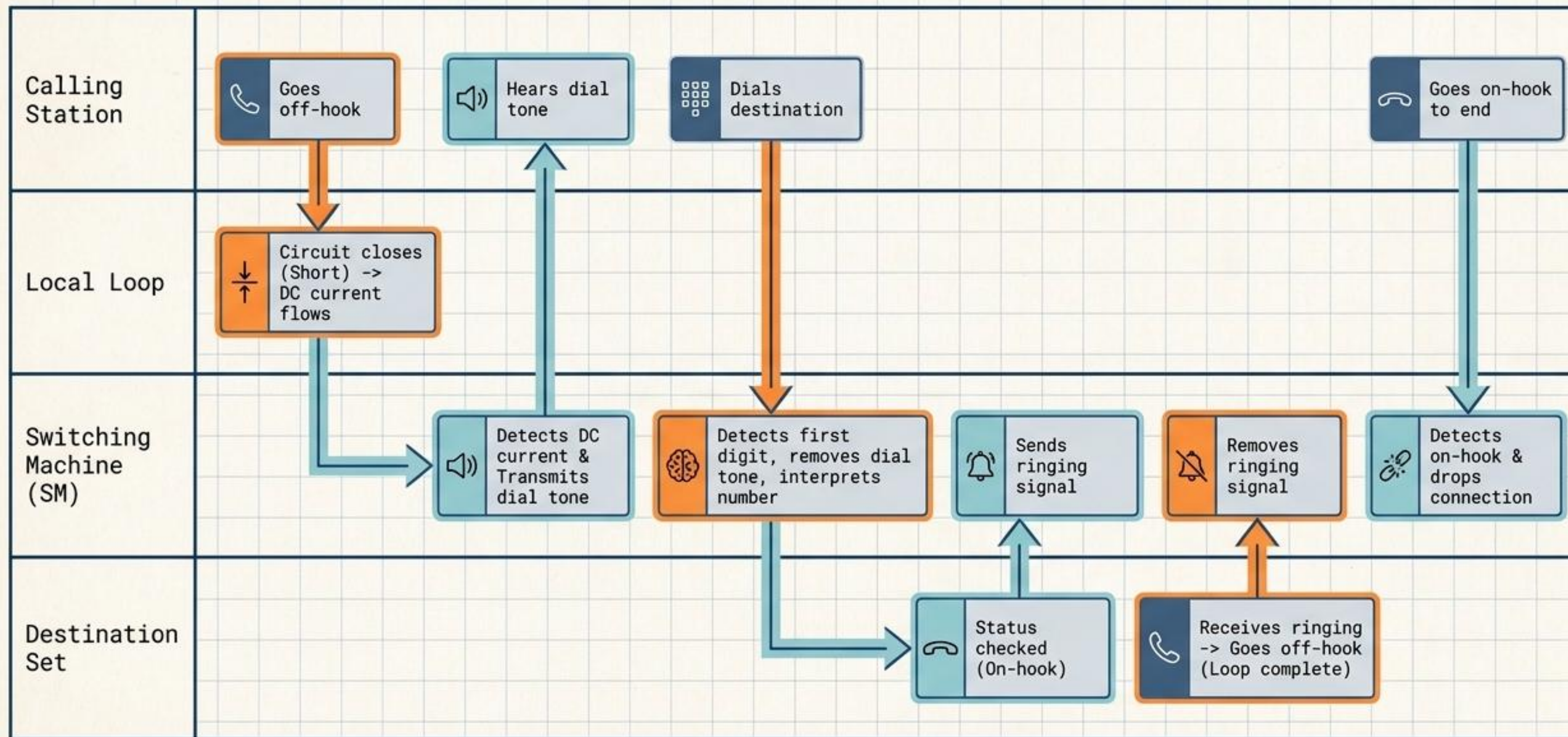


Virtual Private Network (VPN)

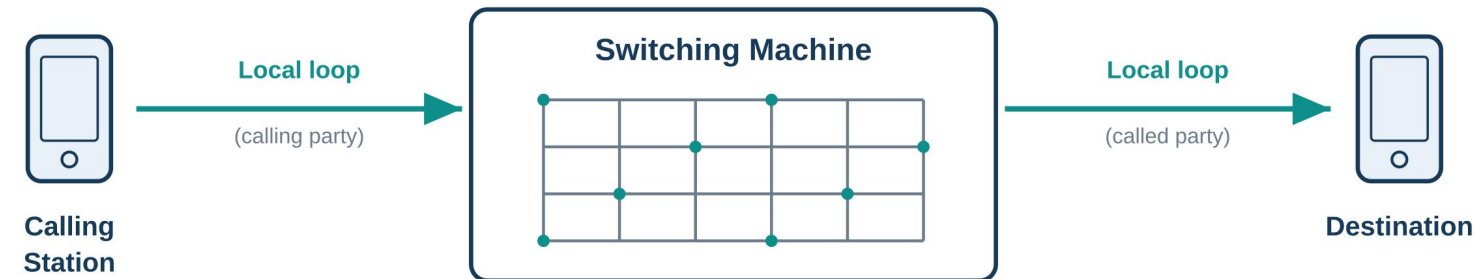


Freephone service

Synthesis: The Lifecycle of a Telephone Call



How a Telephone Call Is Made

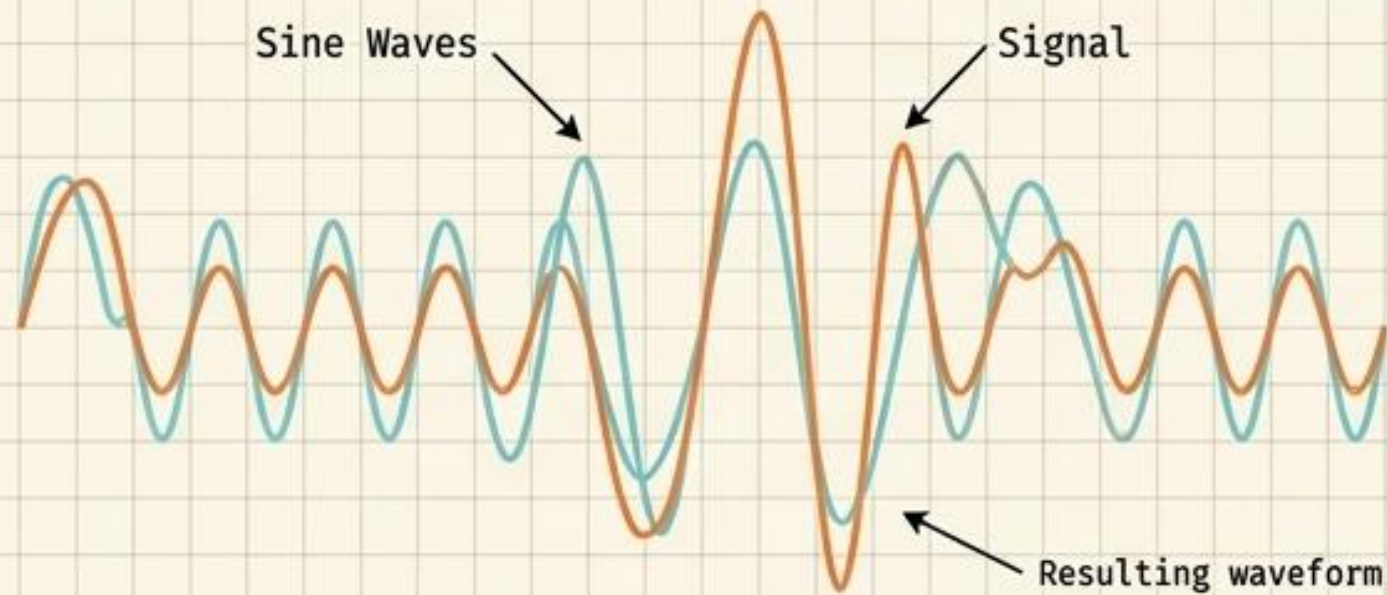


- 1 Calling station goes off-hook.
- 2 Machine detects DC continuity and returns dial tone.
- 3 Caller dials the destination number.
- 4 On the first digit, dial tone is removed.
- 5 Number is interpreted; the destination loop is seized.

- 6 Before ringing, it checks if the far set is on- or off-hook.
- 7 Off-hook → busy signal; on-hook → ringing signal sent.
- 8 Destination answers; the loop completes and DC flows.
- 9 Switching machine removes the ringing signal.
- 10 Either end hangs up → the connection is dropped.

The Modern Superstructure

The Language of Signaling



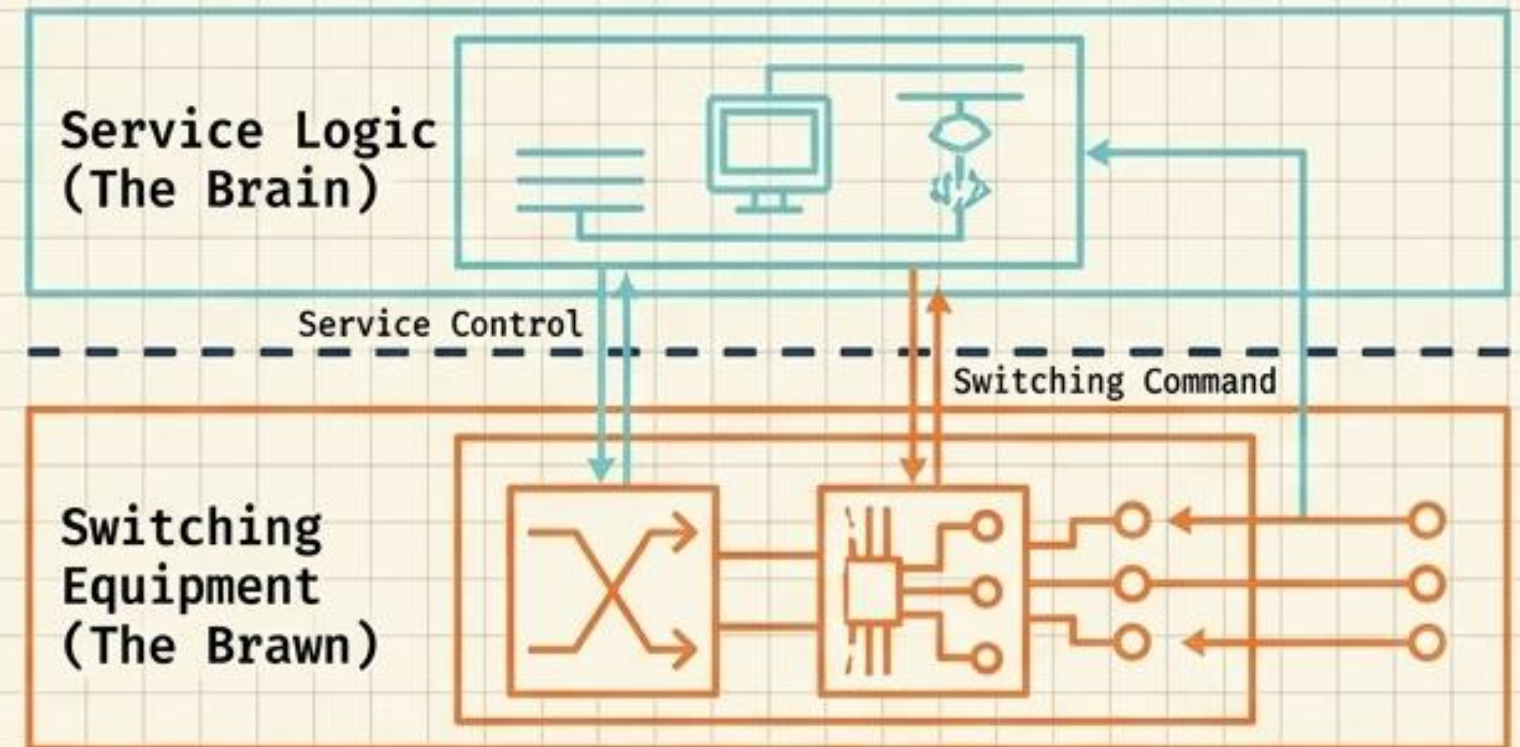
Dial Tone

Not a single sound, but an audible signal comprising two frequencies (350Hz and 440Hz) linearly combined and transmitted simultaneously by the switching machine.

DTMF (Touch Tone)

A highly efficient two-of-eight encoding scheme. Every dialed digit represents the linear addition of two specific frequencies to transfer data instantly to the central office.

The Intelligent Network Architecture



Intelligent Network Benefits & Requirements

- Separates service logic from core hardware.
- Allows new services without redesigning physical switches.
- Enables applications like Pre-paid systems, Calling Cards, and VPN free-phone routing.
- Requires stringent Quality of Transmission (Volume, Frequency Response, Interference).